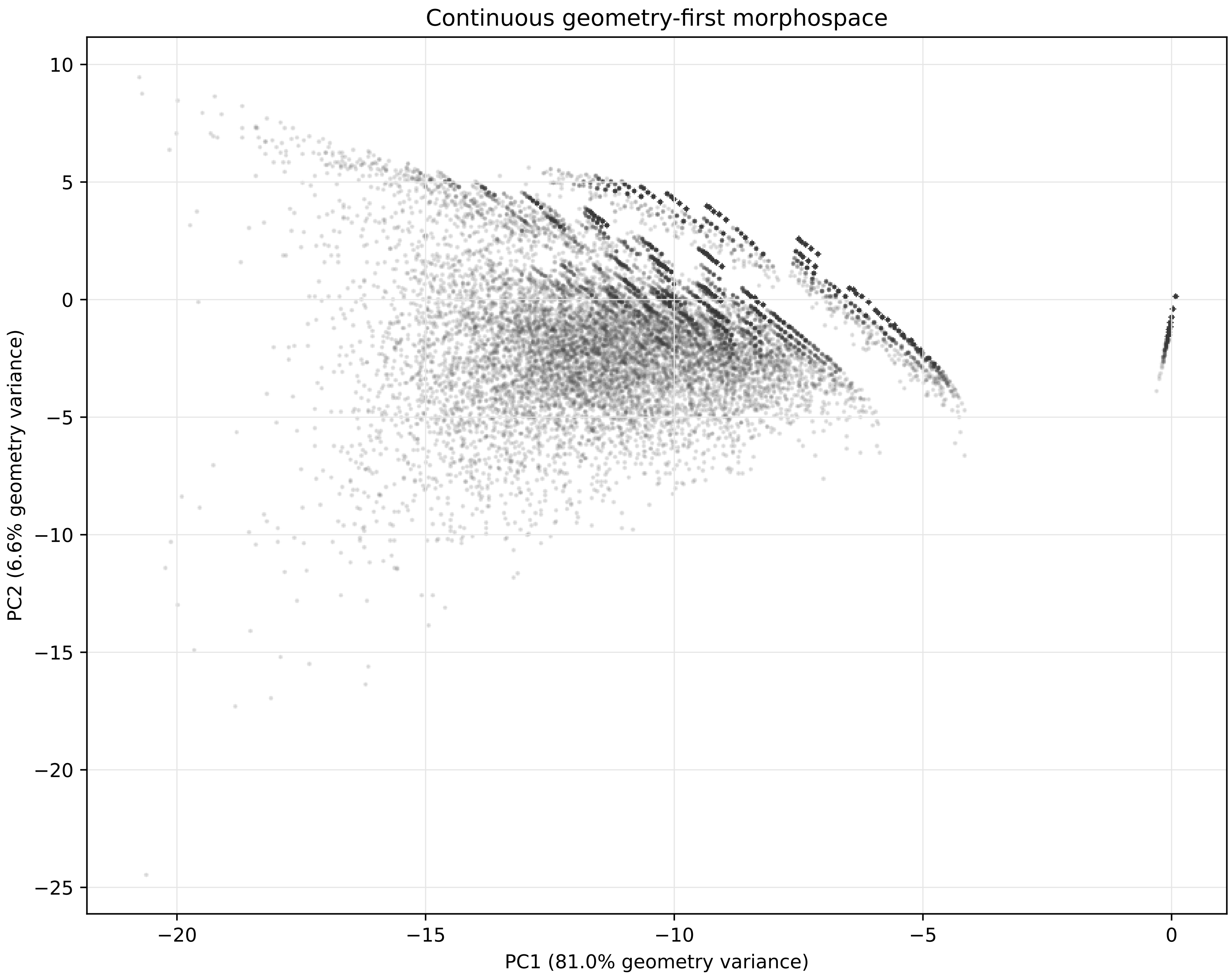
Fire VASE Developmental Morphospace

Geometry defines the morphospace; climate is attached afterward and evaluated as coupling.



Fires represented: 278,569
Climate-complete fires: 237,235
Geometry PC1-PC5 variance: 96.3%
Median duration: 1.0 days
Median final area: 0.43 km2

Interpretation: categories are labels placed inside the space, not the axes of the space. The first pass estimates coupling with linear information proxies, so low R2 should be read as weak linear recoverability rather than proof of no relationship.

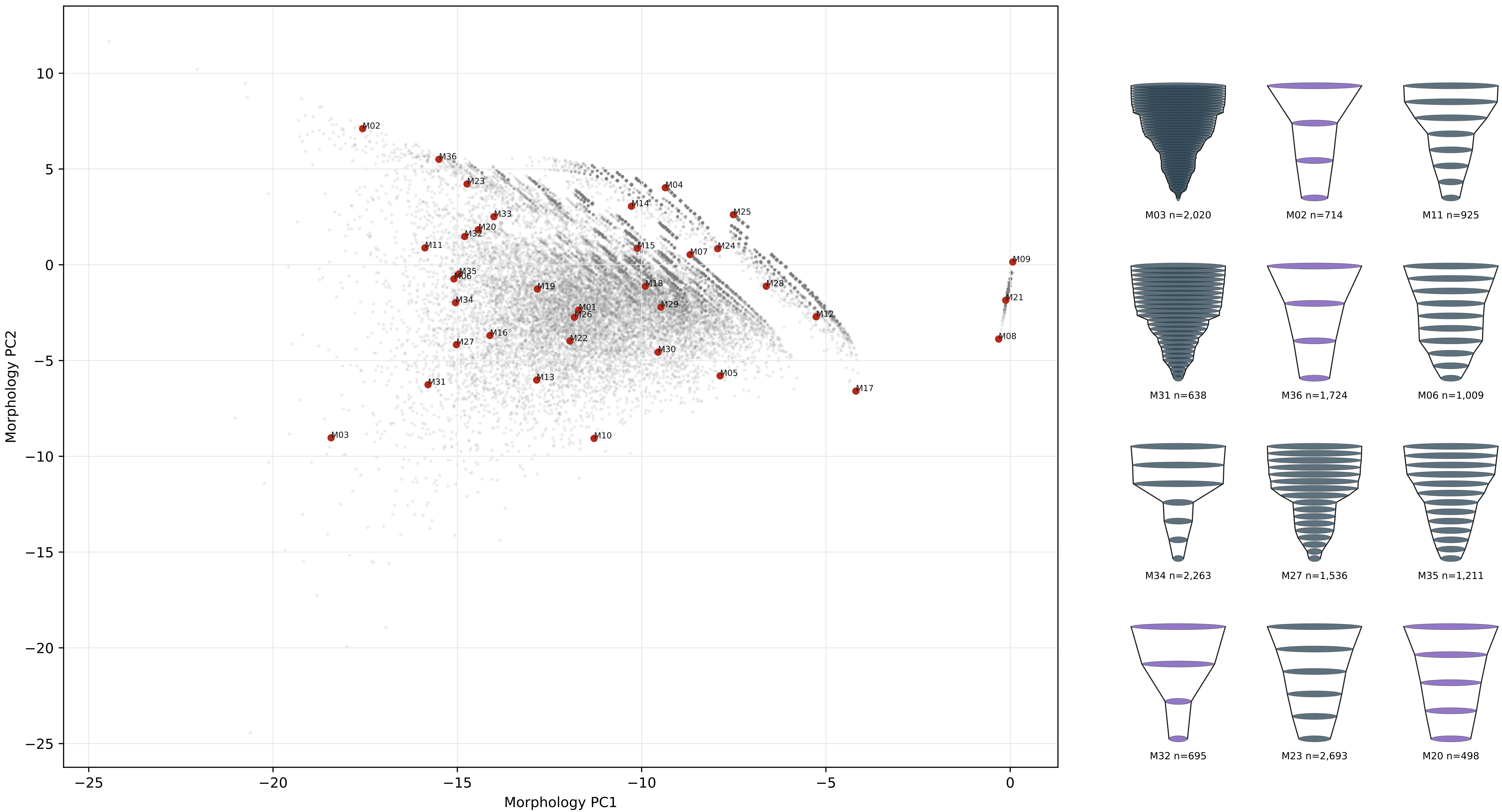
Mean directional coupling proxy

P(climate | morphology): R2=0.020

P(morphology | climate): R2=0.191

Figure 2: Morphospace With Representative Fire VASEs

Medoids are selected by farthest-point coverage in PC1-PC3. Counts are fires assigned to each nearest medoid.



What Dimensions Organize Wildfire Development?

Largest absolute geometry-feature loadings for the first three morphospace axes.

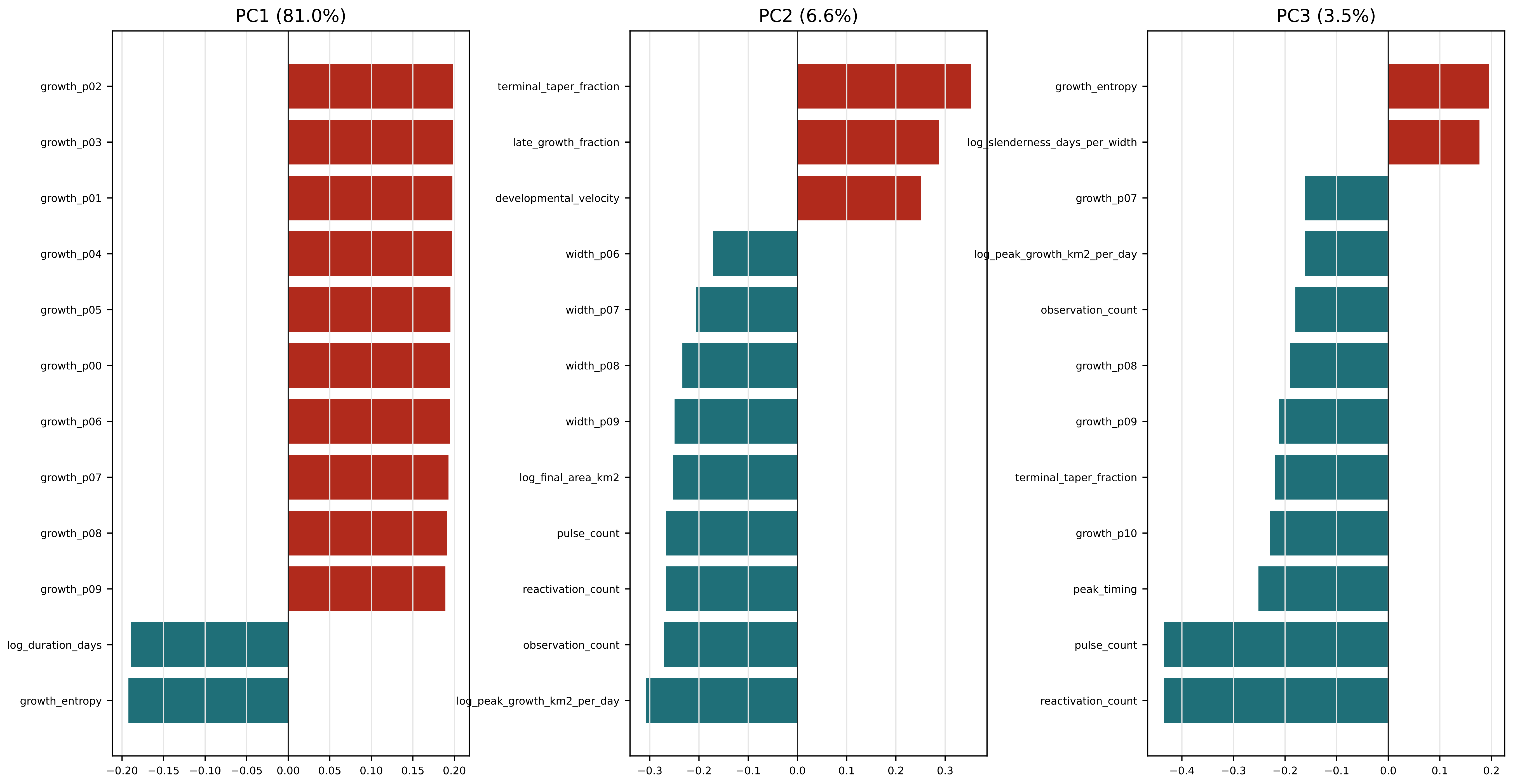


Figure 3: Morphological Field Guide, Page 1

Each representative is a real medoid VASE. Rings are observed developmental slices; colors show maximum temperature where available.

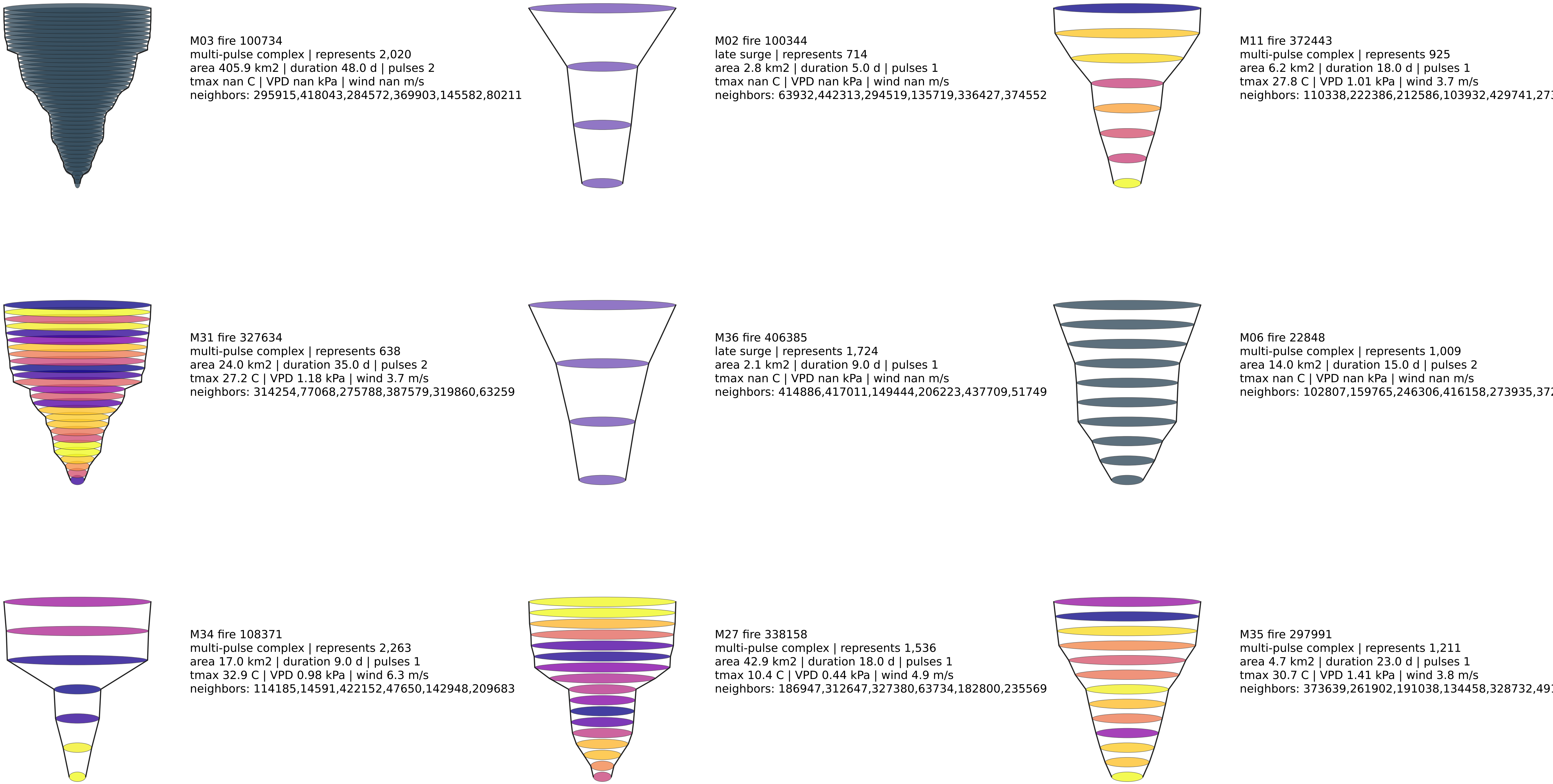
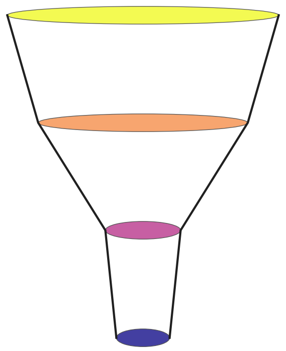
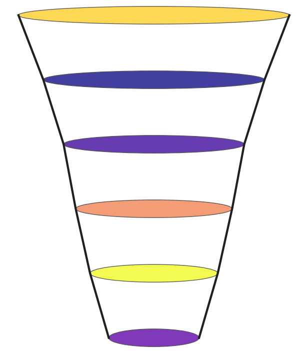


Figure 3: Morphological Field Guide, Page 2

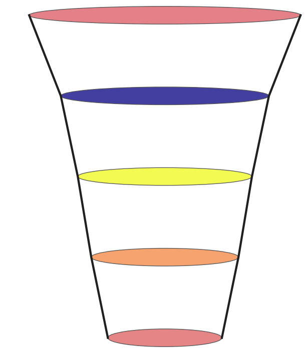
Each representative is a real medoid VASE. Rings are observed developmental slices; colors show maximum temperature where available.



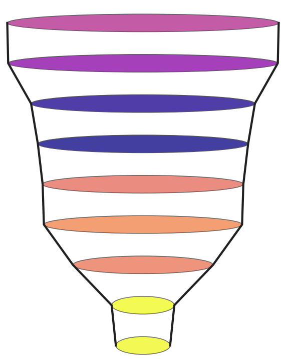
M32 fire 423200
late surge | represents 695
area 11.2 km2 | duration 5.0 d | pulses 1
tmax 18.1 C | VPD 0.68 kPa | wind 5.1 m/s
neighbors: 115473,6053,125993,228423,235216,379999



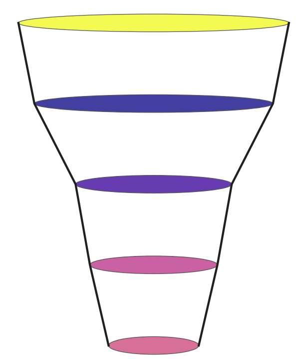
M23 fire 116797
multi-pulse complex | represents 2,693
area 1.9 km2 | duration 14.0 d | pulses 1
tmax 23.8 C | VPD 1.05 kPa | wind 3.9 m/s
neighbors: 28146,52136,111174,437657,57445,46988



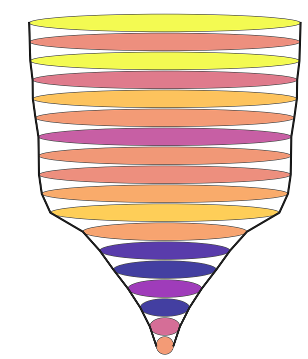
M20 fire 220581
late surge | represents 498
area 3.6 km2 | duration 6.0 d | pulses 2
tmax 25.4 C | VPD 1.51 kPa | wind 2.8 m/s
neighbors: 328305,400100,169709,353839,277949,2113



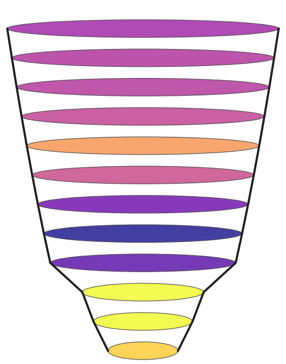
M16 fire 259789
multi-pulse complex | represents 1,153
area 16.1 km2 | duration 9.0 d | pulses 2
tmax 26.1 C | VPD 1.42 kPa | wind 2.0 m/s
neighbors: 321540,308539,272188,55292,168600,46560



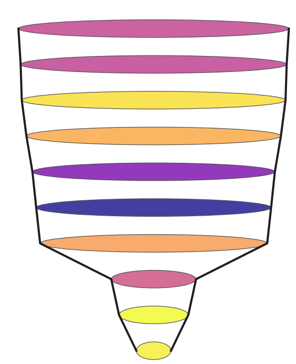
M33 fire 24803
late surge | represents 2,601
area 1.9 km2 | duration 14.0 d | pulses 1
tmax 31.7 C | VPD 0.97 kPa | wind 3.1 m/s
neighbors: 412547,390892,329872,239156,2258,98836



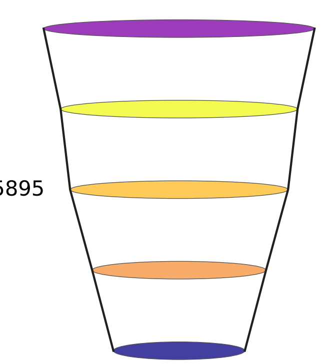
M13 fire 40025
multi-pulse complex | represents 788
area 52.2 km2 | duration 20.0 d | pulses 1
tmax 35.1 C | VPD 1.76 kPa | wind 4.8 m/s
neighbors: 275948,253531,20928,294559,42282,22147



M19 fire 9846
multi-pulse complex | represents 7,297
area 3.2 km2 | duration 22.0 d | pulses 1
tmax 25.1 C | VPD 0.85 kPa | wind 3.3 m/s
neighbors: 25853,46131,394795,300277,7969,149292



M22 fire 328682
multi-pulse complex | represents 3,082
area 13.1 km2 | duration 19.0 d | pulses 1
tmax 23.6 C | VPD 0.83 kPa | wind 4.1 m/s
neighbors: 328916,179082,208058,392042,123615,225895



M26 fire 414357
multi-pulse complex | represents 738
area 3.6 km2 | duration 6.0 d | pulses 3
tmax 33.2 C | VPD 1.41 kPa | wind 3.9 m/s
neighbors: 113743,262821,222589,262063,114797,1269

Figure 3: Morphological Field Guide, Page 3

Each representative is a real medoid VASE. Rings are observed developmental slices; colors show maximum temperature where available.

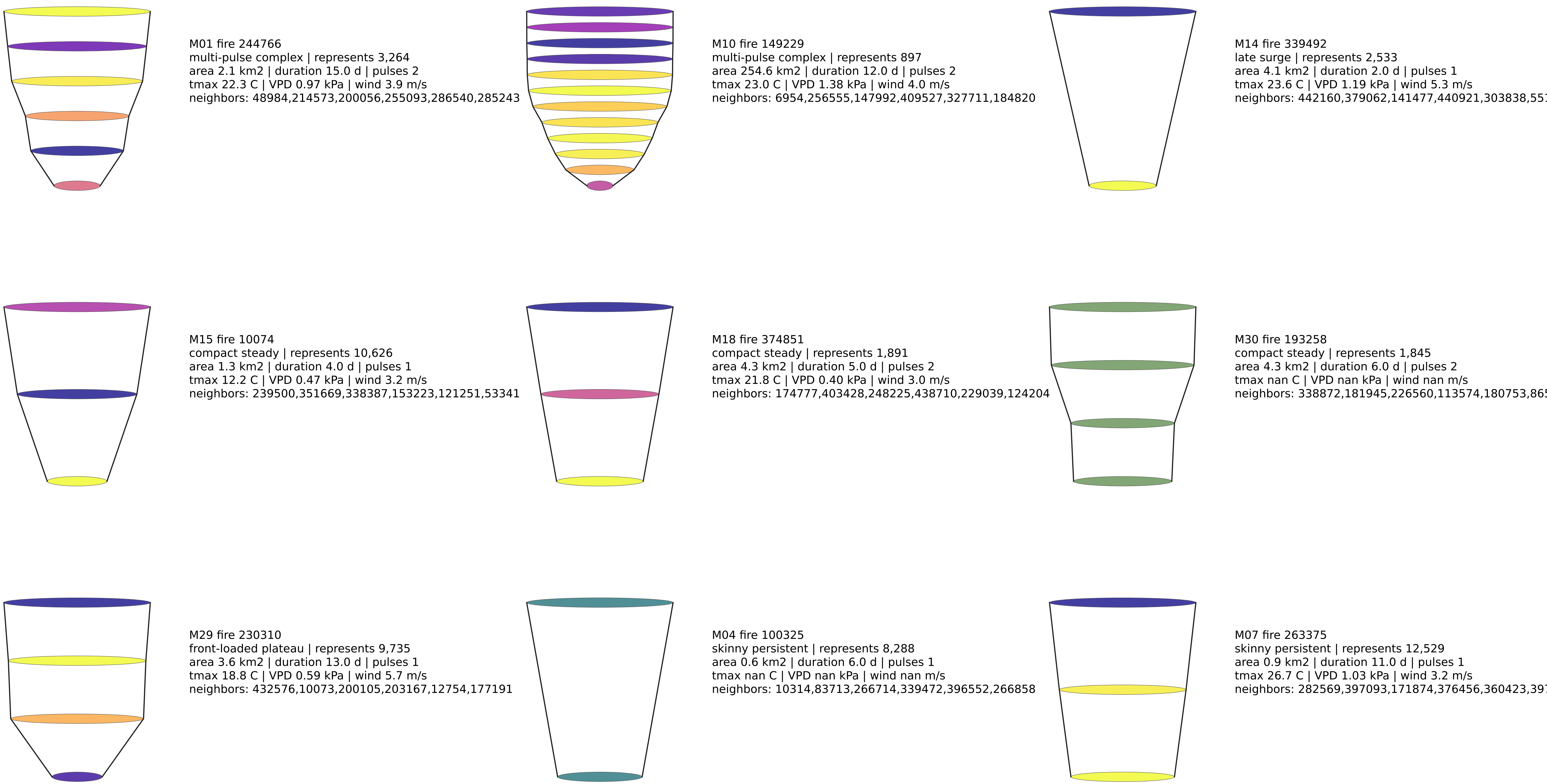
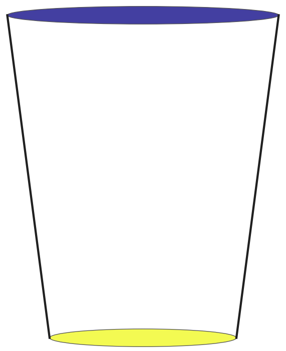
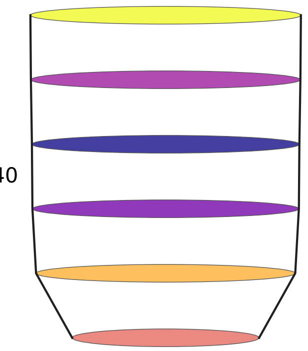


Figure 3: Morphological Field Guide, Page 4

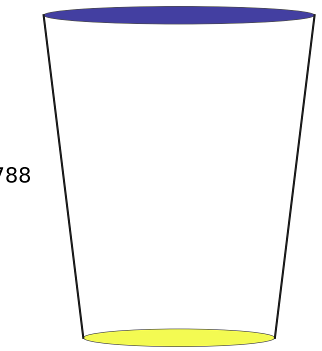
Each representative is a real medoid VASE. Rings are observed developmental slices; colors show maximum temperature where available.



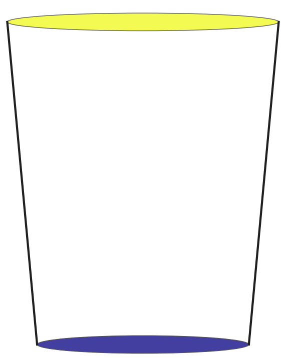
M24 fire 116899
late surge | represents 1,188
area 4.1 km2 | duration 2.0 d | pulses 1
tmax 26.1 C | VPD 1.29 kPa | wind 5.8 m/s
neighbors: 338145,124882,267425,57016,199083,443140



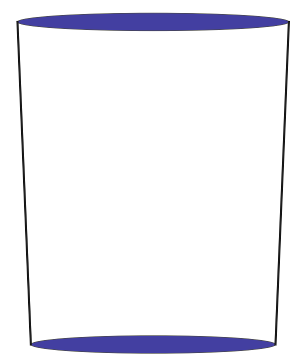
M05 fire 113352
multi-pulse complex | represents 1,216
area 28.8 km2 | duration 7.0 d | pulses 1
tmax 18.1 C | VPD 0.89 kPa | wind 4.6 m/s
neighbors: 427890,97523,144592,239590,317242,254788



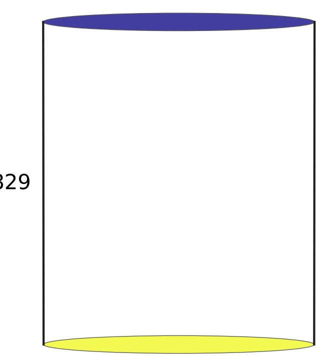
M25 fire 100098
skinny persistent | represents 25,672
area 0.4 km2 | duration 6.0 d | pulses 1
tmax 24.2 C | VPD 1.15 kPa | wind 2.2 m/s
neighbors: 193985,416289,273014,416275,262681,3147



M28 fire 386145
compact steady | represents 1,549
area 4.9 km2 | duration 2.0 d | pulses 1
tmax 34.9 C | VPD 2.60 kPa | wind 4.0 m/s
neighbors: 231068,252403,227714,441976,6356,387890

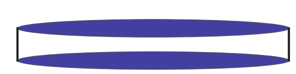


M12 fire 128481
front-loaded plateau | represents 4,629
area 3.4 km2 | duration 2.0 d | pulses 1
tmax 23.9 C | VPD 1.23 kPa | wind 2.7 m/s
neighbors: 225330,387615,39147,184524,236628,293829

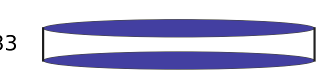


M17 fire 101585
front-loaded plateau | represents 49
area 73.8 km2 | duration 2.0 d | pulses 1
tmax 31.1 C | VPD 2.06 kPa | wind 3.7 m/s
neighbors: 256776,147746,117227,357738,225782,3452

M08 fire 5760
single flash | represents 22
area 37.8 km2 | duration 1.0 d | pulses 1
tmax 22.5 C | VPD 0.79 kPa | wind 2.8 m/s
neighbors: 5629,87148,305887,394586,101580,5832



M21 fire 10398
single flash | represents 6,388
area 2.8 km2 | duration 1.0 d | pulses 1
tmax 23.9 C | VPD 1.11 kPa | wind 4.3 m/s
neighbors: 275924,439427,97824,54368,293710,230633



M09 fire 100
single flash | represents 154,663
area 0.2 km2 | duration 1.0 d | pulses 1
tmax 26.6 C | VPD 0.82 kPa | wind 4.3 m/s
neighbors: 10001,10002,100003,100011,10000,100031

Figure 4: Climate Mapped Onto Morphospace

Climate is evaluated after geometry defines the morphospace.

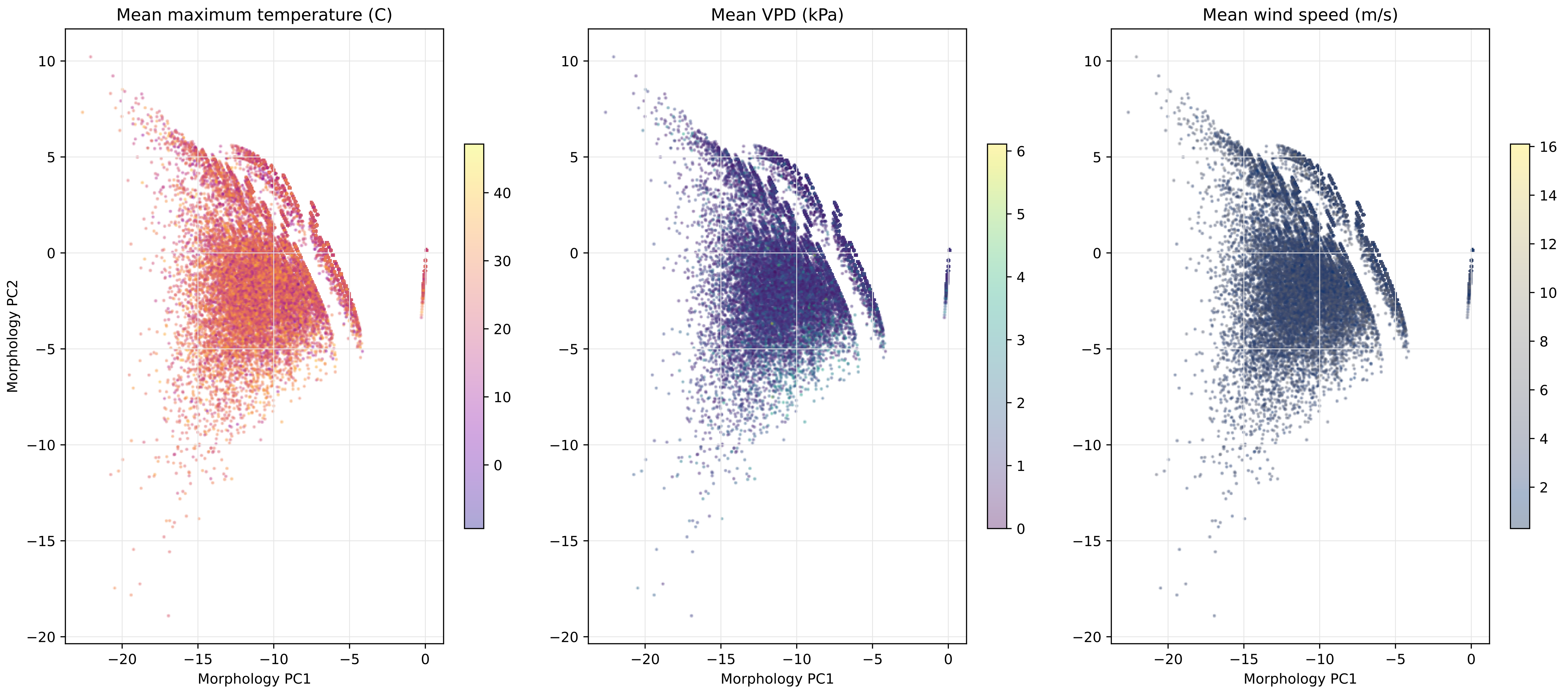


Figure 5: Matched Comparisons

Pairs estimate resilience: similar morphology under different climate, and similar climate with different morphology.

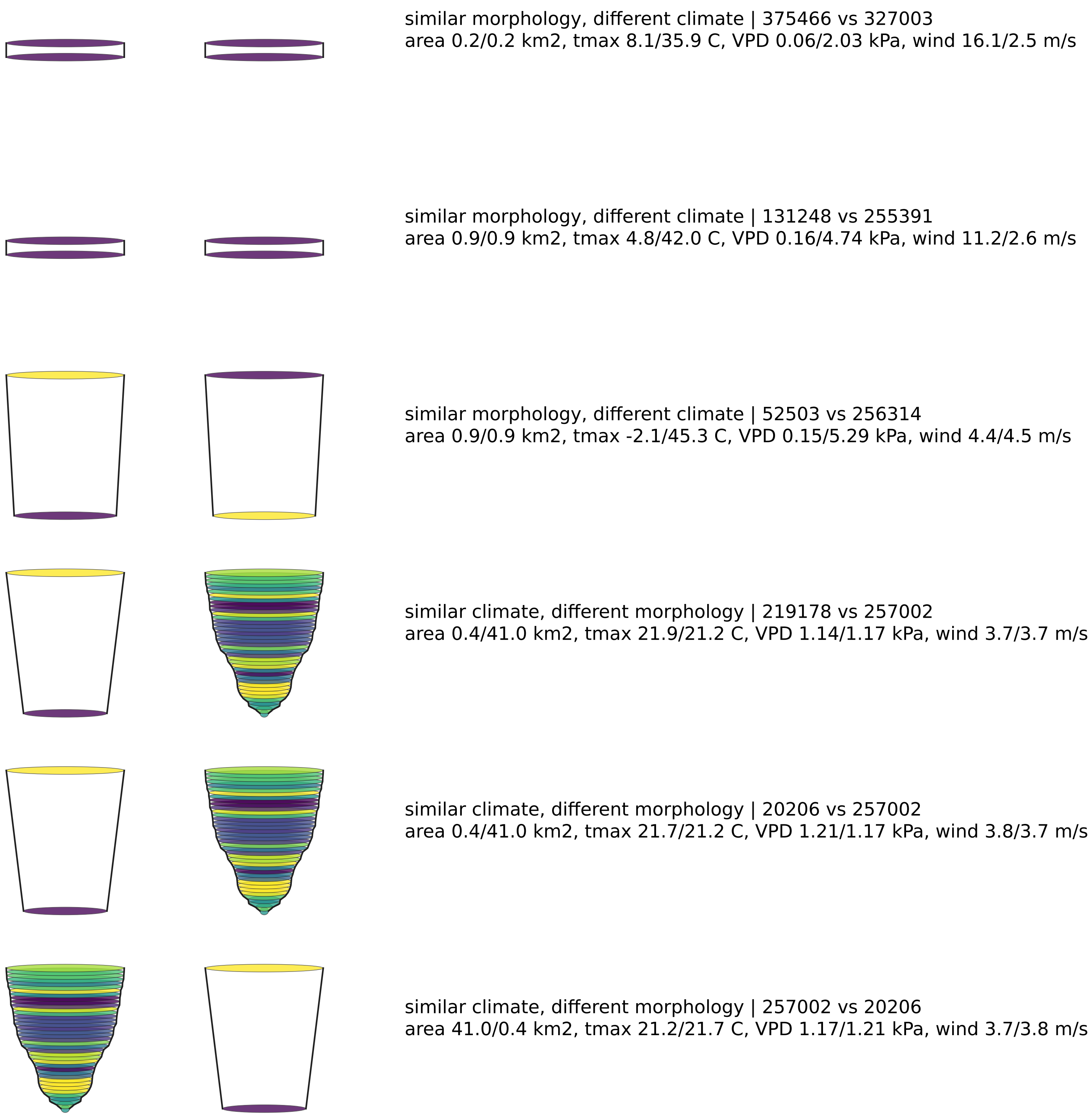


Figure 6: Developmental Control Profile

Predictive information proxies compare climate-only, geometry-only, and combined models through development.